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SECTION 1

ASSIGNMENT 1 - TECHNICAL REPORT

TITLE:

**TOPIC 19: ARTIFICIAL INTELLIGENT AND MACHINE LEARNING DATA
PIPELINES**

GROUP 6

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EXECUTIVE SUMMARY

This report discusses the use of Artificial Intelligence (AI) and Machine Learning (ML) in modern data pipelines. The primary goal of this report is to demonstrate how AI and ML enhance the data handling process, from data collection to decision-making. The report also explains the advantages and challenges of AI/ML data pipelines, as well as their practical uses.

Based on the discussion, AI and ML can make data pipelines work better by automating tasks like cleaning, transforming and monitoring data. This reduces human error and speeds up the data processing process. For example, in a fraud detection system, the AI pipelines are able to analyze large amounts of data and make detection more accurate. However, there are still a lot of challenges like the high cost of implementation, the technical difficulties, the lack of skilled workers and issues related to data privacy and transparency.

To overcome these challenges, organizations should work on better data management, use MLOps practices and switch to cloud platforms for better scalability. In conclusion, AI and ML data pipelines are important in modern systems but they need to be planned well and improved all the time to work well.

1.0 INTRODUCTION

The rapid development of digital technology has completely changed the way enterprises collect, process and use data. Integrating machine learning (ML) and artificial intelligence (AI) into the data pipeline system is one of the most important advances. Traditional data pipelines are mainly based on rules and static, and the purpose is to transmit data from the source system to the storage device, but it lacks intelligence. However, with the increase in data volume, data speed and data type, the shortcomings of these systems are also increasingly prominent, such as slow processing speed, error-prone manual conversion, and lack of flexibility to cope with changing data patterns.

Although artificial intelligence and machine learning has many advantages, enterprises will still face many challenges, such as the high cost of implementation, the shortage of qualified personnel, data governance problems, and the requirements of model openness. This report will discuss the architecture, use, advantages and difficulties of artificial intelligence and machine learning data pipeline, and put forward suggestions for enterprises to successfully apply this technology.

2.0 BACKGROUND / LITERATURE REVIEW

Machine Learning (ML) and Artificial Intelligence (AI) have become important parts of modern data pipeline architectures. They make it possible to process, analyze and gain insights from data quickly and easily. Prior research indicates that AI-driven automation rapidly reduces manual tasks, including data cleaning, transformation and integration, resulting in enhanced efficiency and minimized human error (Ahmed et al., 2025; Chandrashekar et al., 2024). Automated ML pipelines have shown that they can improve data accuracy and make complicated analytical processes easier in fields like the study of genetics.

Recent developments have also highlighted the importance of cloud and edge computing in facilitating scalable and real-time data processing. Cloud platforms help businesses handle large amounts of data and quickly set up machine learning models. Moreover, the edge computing lets data processing happen closer to where the data is (Guntupalli, 2025). MLOps has also changed the way DevOps is done by adding continuous integration, deployment and monitoring just for machine learning models.

Hyperautomation is getting a lot of attention as a new trend because it combines AI, ML and robotic process automation (RPA) to make systems that work on their own. Also, multi-source data fusion and explainable AI are becoming very important areas of research as companies try to make models more accurate while also making sure they are transparent and reliable (Lu & Cheng, 2025).

Even with these improvements, there are still gaps in the research. Data quality and governance remain significant challenges, as biased or incomplete datasets may significantly affect model performance (Mishra & Darade, 2024). Also, high infrastructure costs and problems with scalability make it hard for smaller businesses to adopt it. Ongoing problems also come from ethical issues and rules that must be followed, like data privacy compliance.

In conclusion, while AI and ML have considerably enhanced data pipelines, further research is required to solve difficulties related to data quality, scalability and ethical considerations, ensuring more secure and dependable systems in the future.

3.0 ANALYSIS AND DISCUSSION

The integration of artificial intelligence and machine learning into traditional data pipelines resulted in an end-to-end automated system that flows data from raw collection to model training and deployment which we called it as Artificial Intelligence and Machine Learning Data Pipeline (AI/ML Data Pipelines) (IBM, 2026; Gubitosa, 2025). This section will discuss how it works along with its case examples, advantages and disadvantages and challenges and future trends.

3.1 How AI/ML Data Pipelines Work

This pipeline is an integration of machine learning and artificial intelligence into a data pipeline to automate complex processes and enable real-time data processing which allow faster

data-driven decision making (Gubitosa, 2025). Thus, there are two key parts in the pipeline which are data processing and machine learning modelling and deployment.

According to Jandhyala (2024), there are four main stages in the data processing part, which are data ingestion, cleaning and preprocessing, transformation and feature engineering, storage and loading. During these stages, raw data is imported into the pipeline from various sources such as NoSQL databases, relational databases, and real-time streaming sources (e.g. IoT devices), which will be examined for any noise, inconsistencies, missing values and duplicates, which are then removed, corrected and handled accordingly. The data also goes format standardization to ensure the data is structured and usable for model training. Then, the cleaned data will undergo transformation including scaling and encoding and will be combined using feature engineering to ensure the data is usable for machine learning algorithms used in the pipeline. Lastly in this part, the processed data and engineered features are stored in a data lake for easy access for AI models.

Into the next part of the pipeline, Domo (2026) mentioned the last two stages are model training and inference and monitoring and feedback loops. The prepared data from before is used to train models or generate predictions. The output of this stage may be used for creating a business dashboard for decision-making. Monitoring and feedback loops are for tracking the model performance, which consist of a three-layer monitoring framework: data layer, feature layer and prediction layer. Each layer monitors a different aspect of the pipeline. For instance, data layer is for data health tracking which includes monitoring data completeness and structural changes which allows any data changes or issues to be detected early before impacting the model. This framework allows catching any arising problem or error in each layer to avoid impacting business outcomes and ensure the model evolves alongside the data.

3.2 Real-World Applications of AI/ML Data Pipelines

Saini et al. (2025) discuss banking fraud detection pipelines where institutions process large volumes of data across markets, products, and transactions while adhering to strict regulatory requirements. A unified AI system was deployed to automate data quality control across the pipeline from ingestion to model outputs using rule-based, statistical, and AI-based methods to detect anomalies and ensure data reliability. The system showed significant improvements, reducing false alerts by approximately five times and improving fraud detection accuracy to 90%.

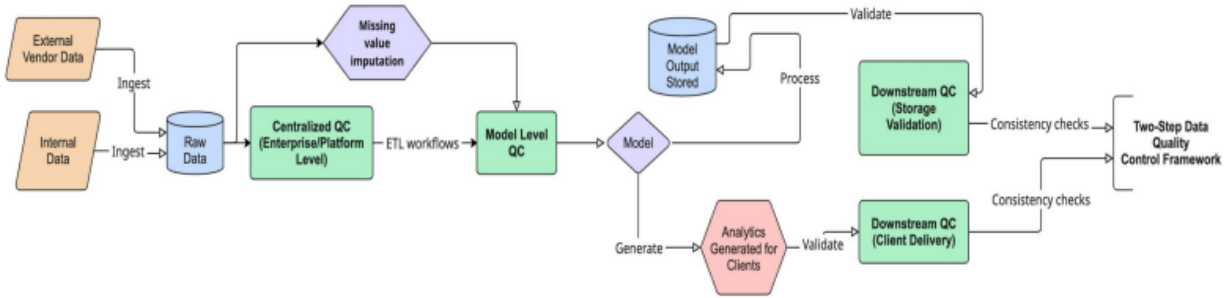


Figure 1: Two-Step Data QC Framework by Saini et al. (2025).

3.3 Advantages and Disadvantages of AI/ML Data Pipeline

There are many advantages of this technology mentioned by Reddy and Adapa (2025) such as reducing human intervention while improving accuracy and throughput. Additionally, these pipelines significantly reduce integration timeframes and improve data quality, providing organisations with enhanced operational efficiency and a competitive advantage in data-driven environments.

However, this technology also has its own disadvantages. Reddy and Adapa (2025) stated that some of the disadvantages of AI/ML data pipelines are the requirement of higher initial technical complexity compared to traditional approaches. Furthermore, this technology requires data science and ML expertise which some organisations may lack.

3.4 Challenges and Future Trends

According to Reddy and Adapa (2025), the challenges of this technology is that it requires specialized explainability components to maintain the transparency of the Artificial Intelligence decisions or prediction. Other than that, this technology may also require significant computational resources during training phases which may cause reduction in performance if not met. However, the future of AI/ML data pipelines points toward greater autonomy and intelligence if developed continuously. Reddy and Adapa (2025) highlighted the emergence of self-healing pipelines that can detect, diagnose, and fix issues with minimal human intervention. Furthermore, energy-efficient pipelines will also emerge in considerations of sustainability to minimize the environmental impact of data integration operations.

4.0 RECOMMENDATIONS

Based on the analysis and discussion in previous sections, several practical recommendations are proposed for organizations seeking to implement or enhance their AI and ML data pipelines.

Before deploying AI or ML processes, enterprises should put the establishment of a strong data governance structure as a priority. Enterprises must adopt systematic data verification, data tracking and anomaly detection at every stage of the process, because data quality is a key determinant of model performance (Saini et al., 2025). By implementing policy-as-code

technology, deviation detection, data traceability and compliance standards can be continuously and effectively implemented. This technology can automatically execute governance rules and integrate them directly into the machine learning operation and maintenance (MLOps) workflow (Mishra & Darade, 2024).

Furthermore, enterprises need to establish a standardized machine learning operation and maintenance (MLOps) architecture to bridge the gap between model development and production deployment. Research shows that up to 85% of machine learning models eventually fail to be put into production due to lack of a clear operation process. By establishing a continuous integration and deployment pipeline tailored to machine learning workloads and strict version control of data sets, model parameters and pipeline configurations, the deployment failure rate can be significantly reduced and regulatory audits can be supported (Reddy & Adapa, 2025).

Next, enterprises should increase investment in cross-functional cooperation to overcome the problem of limited internal expertise. Coordination between multiple roles such as data scientists, data engineers, field experts and business stakeholder is crucial to the successful implementation of artificial intelligence/machine learning processes (Reddy & Adapa, 2025). By forming a professional artificial intelligence team with clear responsibilities and establishing a shared document process, we can narrow the knowledge gap and ensure that process development is consistent with strategic business goals.

In addition, enterprises should also use cloud-native platforms to solve infrastructure costs and scalability problems, because the cloud environment provides a pay-as-you-go pricing model, which can not only reduce early capital investment, but also provide the computing flexibility required to manage different workloads in the intensive model training stage (Domo, 2026). In addition, platforms such as Google Vertex AI and Azure Machine Learning integrate MLOps tools to speed up pipeline deployment while ensuring enterprise-level security.

5.0 CONCLUSION

This report has discussed the technology of artificial intelligence and machine learning data pipelines along with its benefits and challenges. These studies highlighted the integration of artificial intelligence and machine learning into data pipelines has resulted in increased efficiency of data processing, analysis, and insight generation and the AI-driven automation helps reduce human errors (IBM, 2026; Gubitosa, 2025). The technology is divided into two main parts which are data processing and machine learning modelling and deployment. The data processing part covers data ingestion, cleaning, transformation, and storage, while the machine learning part handles model training, inference, and continuous monitoring to ensure reliable predictions (Jandhyala, 2024; Domo, 2026).

A real-world application, banking fraud detection was presented where AI/ML data pipelines were used to automate data quality control to detect anomalies and ensure data reliability which resulted in a fivefold reduction of false alert and 90% fraud detection accuracy (Saini et al., 2025). Although AI/ML data pipelines can improve accuracy and throughput, it requires data science and ML expertise to implement effectively and specialized explainability components for transparency (Reddy & Adapa, 2025). These disadvantages and challenges highlight the importance of planning before adoption. Thus, organizations are recommended to establish strong data governance and invest in skills development and teamwork. Looking ahead, the emergence of self-healing pipelines and energy-efficient systems suggests that AI/ML data pipelines will continue evolving toward greater autonomy, becoming increasingly essential in modern enterprise data architectures.

REFERENCES

- Ahmed, K. R., Islam, M. S., Khan, M. T., Mehbuba, M. S., Apu, B. D., Noyeem, M. M. I., & Goyal, S. K. (2025, December). DevOps for AI: Challenges and Opportunities in Integrating Machine Learning Models into Continuous Deployment Pipelines. In *2025 7th International Symposium on Advanced Electrical and Communication Technologies (ISAECT)* (pp. 1-6). IEEE.
- Chandrashekar, K., Niranjana, V., Vishal, A., & Setlur, A. S. (2024). Integration of artificial intelligence, machine learning and deep learning techniques in genomics: review on computational perspectives for NGS analysis of DNA and RNA seq data. *Current Bioinformatics*, 19(9), 825-844.
- Domo. (2026, March 31). *AI data pipelines: What they are and how to build one*. Domo. <https://www.domo.com/glossary/ai-data-pipelines#how-ai-data-pipelines-work>
- Gubitosa, B. (2025, May 28). *AI Data Pipeline: A Comprehensive Guide*. Riverty. <https://riverty.io/data-learning-center/ai-data-pipeline/>
- Guntupalli, R. (2025, June). AI-Powered Data Analytics in Cloud Computing. In *International Conference on Data Analytics & Management* (pp. 280-289). Cham: Springer Nature Switzerland.
- IBM. (2026, January 21). *What is a machine learning pipeline?* IBM. <https://www.ibm.com/think/topics/machine-learning-pipeline>
- Jandhyala, V. S. V. (2024). Mastering Data Pipelines for AI: A Beginner's Guide to Building Efficient Workflows. *International Journal For Multidisciplinary Research*, 6(5). <https://doi.org/10.36948/ijfmr.2024.v06i05.29550>
- Lu, H., & Cheng, Y. F. (2025). Artificial intelligence in energy pipelines: opportunities and risks. *Engineering*.

Mishra, V., & Darade, H. (2024, December). Harnessing Big Data: An Expert Review of Trends, Challenges, and Future Directions with a Focus on Data Quality. In *2024 Third International Conference on Artificial Intelligence, Computational Electronics and Communication System (AICECS)* (pp. 1-6). IEEE.

Reddy, N. R. S., & Adapa, M. (2025). AI-Driven Data Integration: Transforming Enterprise Data Pipelines through Machine Learning. *Journal of Computer Science and Technology Studies*, 7(12), 110-119.

Saini, D., Jain, B., Ujjwal, N., Sommer, P., Mbanga, D. R., & Mehta, D. (2025). A Unified AI System For Data Quality Control and DataOps Management in Regulated Environments. *arXiv preprint arXiv:2512.05559*.

APPENDIX / LOGBOOK

Attach your record of the research process, including AI prompts used (if any), verification steps, keywords, and tools consulted. Screenshots from GitHub or logbook entries can be included here.

1. AI Prompts Used				
No.	Date	Tool	Prompt Asked	Purpose
1	12/4/2026	Claude (Anthropic)	"What is an AI/ML data pipeline?"	Understanding basic concept
2	12/4/2026	Claude (Anthropic)	"What part can be included in analysis and discussion?"	Report structuring
3	12/4/2026	Claude (Anthropic)	"Find references that say AI/ML pipeline is two pipelines combined"	Finding supporting references
4	12/4/2026	Claude (Anthropic)	"Give me APA format for this reference"	Reference formatting
5	12/4/2026	Claude (Anthropic)	"Check my paragraph for grammar and accuracy"	Writing verification
6	12/4/2026	SCISPACE	"can you find related references to stages in Artificial Intelligence and Machine Learning Data Pipelines and combination of machine learning pipelines and data pipelines, with integration of artificial intelligence"	Finding supporting references
7	12/4/2026	SCISPACE	"find article related to real-world applications or case example of Artificial Intelligence and Machine Learning Data Pipelines"	Finding supporting references

Figure 2: AI Prompt Used By Lubna Al Haani

No.	Date	Tools Used	Prompt Asked	Purpose
1	09/04/2026	Scopus, Google Scholar	Artificial Intelligence and Machine Learning data pipelines journal articles 2023–2026	To search for relevant academic papers and understand the topic.
2	09/04/2026	ChatGPT, Scopus	Explain AI and ML data pipelines in simple terms	To understand the basic concept before writing.
3	11/04/2026	ChatGPT, Claude	Summarize AI and ML data pipelines for literature review	To gather main ideas for the Background / Literature Review section.
4	11/04/2026	ChatGPT, Claude	Explain current trends in AI and ML data pipelines	To include updated information in the literature review.
5	11/04/2026	ChatGPT, Claude	Explain challenges of AI and ML data pipelines	To identify issues to include in the literature review.
6	12/04/2026	ChatGPT	Check if there is any grammatical error in my sentences and improve them	To ensure the writing is clear and grammatically correct.
7	12/04/2026	ChatGPT	Check if there is any grammatical error in my sentences and improve them	To improve clarity and match the report writing style.

Figure 3: AI Prompt Used By Nurul Asyikin

No.	Date	Tools Used	Prompt Asked	Purpose
1	03/04/2026	Scopus	Artificial Intelligent and Machine Learning Data Pipelines	Gather relevant academic materials
2	10/04/2026	ChatGPT, Scopus, Claude	Artificial Intelligent and Machine Learning Data Pipelines	Compare and understand the topic.
3	10/04/2026	ChatGPT, Claude	What are AI and ML data pipelines and how do they work.	Get more Understanding of the topic.
4	12/04/2026	ChatGPT, Claude	Recommendation AI and ML	Review the best recommendation for the organization on AI/ML data pipeline.
5	12/04/2026	ChatGPT	Shorten and simplify introduction and recommendation sections	To make the report more concise and readable

Figure 4: AI Prompt Used By Anis Safiya